

$$\begin{array}{ccc}
 \mathcal{I}(U) & \xrightarrow{\hat{f}} & \mathcal{I}(U) \\
 \varphi \downarrow & & \downarrow \varphi \\
 S^1 & \xrightarrow{z^d} & S^1
 \end{array}$$